

Purshottapatnam Rakesh

NLP Engineer | Telangana, India

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PROFESSIONAL SUMMARY

Experienced NLP Engineer with a track record of building multilingual NLP pipelines supporting Hindi and regional Indian languages. Strong expertise in tokenisation, embeddings, and fine-tuning for domain-specific corpora.

TECHNICAL SKILLS

Tools & Technologies: Pandas, NumPy, VS Code, GitHub

Programming Languages: Python, R, SQL

ML Libraries & Frameworks: LightGBM, TensorFlow, Keras, Pandas

Deep Learning & Neural Networks: CNN, Transfer Learning, LSTM, Neural Networks

Soft Skills: Team Collaboration, Time Management, Analytical Thinking, Leadership, Communication, Problem Solving

PROJECTS

Medical Image Classification using CNN

Tech Stack: TensorFlow, CNN, NumPy, Python

- Built a ResNet-50 transfer learning model for pneumonia detection from chest X-rays.
- Trained on NIH dataset (10K images), achieving 94% accuracy and 0.92 AUC.
- Implemented Grad-CAM heatmaps for model interpretability and clinical validation.

Real-Time Object Detection for Smart Surveillance

Tech Stack: Python

- Trained a YOLOv8 model on a custom 15K-image dataset for multi-class object detection.
- Achieved mAP@50 of 0.87 with real-time inference at 30 FPS on edge devices.
- Integrated OpenCV pipeline for video stream processing and alert generation.

Big Data Customer Segmentation Pipeline

Tech Stack: Python, Pandas

- Built a Spark-based ETL pipeline processing 500GB of daily transactional data.
- Applied K-Means and DBSCAN clustering to segment 2M+ customers into 8 cohorts.
- Automated pipeline scheduling using Apache Airflow with email alerting on failure.

EDUCATION

B.Tech – Data Science

KASIREDDY NARAYAN REDDY COLLEGE OF ENGINEERING AND RESEARCH | 2025 | CGPA: 7.37

Class XII – TELANGANA STATE BOARD OF INTERMEDIATE EDUCATION

2021 | 9.22%

Class X – BOARD OF SECONDARY EDUCATION

2019 | 9.2%